SSH BRUTE FORCE ATTACK - CROSS-PLATFORM DETECTION LAB

DETAILED INCIDENT REPORT

EXECUTIVE SUMMARY

This lab demonstrates a simulated SSH brute force attack against a Windows system and documents detection from both the attacker perspective (Linux/Kali) and the victim perspective (Windows).

The attack successfully compromised the vbox user account through password guessing, followed by post-compromise reconnaissance. All activity was captured and analyzed using cross-platform logging mechanisms.

KEY FINDINGS:

✓ 6 SSH authentication attempts detected

✓ 5 failed login attempts (brute force pattern)

✓ 1 successful login (account compromised)

✓ 3 post-compromise reconnaissance commands executed

✓ Attack completed in under 30 seconds

ATTACK TIMELINE

OVERVIEW

Time	Event	Source	Details
T+0s	SSH Brute Force Starts	Kali (192.168.56.103)	Attacker initiates

T+2s	Failed Attempt 1	Windows Event Log	Event ID 4625
T+4s	Failed Attempt 2	Windows Event Log	Event ID 4625
T+6s	Failed Attempt 3	Windows Event Log	Event ID 4625
T+8s	Failed Attempt 4	Windows Event Log	Event ID 4625
T+10s	Failed Attempt 5	Windows Event Log	Event ID 4625
T+12s	SUCCESSFUL LOGIN	Windows Event Log	Event ID 4624
T+13s	Reconnaissance 1	Windows Event Log	Event ID 4688
T+15s	Reconnaissance 2	Windows Event Log	Event ID 4688
T+17s	Reconnaissance 3	Windows Event Log	Event ID 4688

DETAILED TIMELINE

PHASE 1: BRUTE FORCE ATTACK (T+0s to T+10s)

Attack Command (From Kali):

```
for i in {1..5}; do
  sshpass -p "wrongpass$i" ssh -l vbox 192.168.56.102 "exit" 2>/dev/null &
done
```

Attacker's Log (ssh_attack_log.txt):

```
=== SSH Brute Force Simulation ===
Start Time: [YOUR_START_TIME]
[TIME_1] Attempt 1: sshpass -p 'wrongpass1' ssh -l vbox 192.168.56.102 'exit'
Permission denied, please try again.
[TIME_2] Attempt 2: sshpass -p 'wrongpass2' ssh -l vbox 192.168.56.102 'exit'
Permission denied, please try again.
[TIME_3] Attempt 3: sshpass -p 'wrongpass3' ssh -l vbox 192.168.56.102 'exit'
Permission denied, please try again.
```

[TIME_4] Attempt 4: sshpass -p 'wrongpass4' ssh -l vbox 192.168.56.102 'exit'

Permission denied, please try again.

[TIME_5] Attempt 5: sshpass -p 'wrongpass5' ssh -l vbox 192.168.56.102 'exit'

Permission denied, please try again.

Victim's Logs (Windows Event Viewer):

Event ID 4625 (Failed Logon) - Entries 1-5:

Event ID: 4625

Computer: DESKTOP-91DJV8I

Account Name: vbox

Source Network Address: 192.168.56.103

Failure Reason: Unknown user name or bad password

Status: 0xC000006D

Time: [TIMESTAMP_1]

Logon Type: 3 (Network)

[Repeat for attempts 2-5 with different timestamps]

PHASE 2: SUCCESSFUL COMPROMISE (T+12s)

Attack Command (From Kali):

sshpass -p "[CORRECT_PASSWORD]" ssh -l vbox 192.168.56.102 "whoami"

Attacker's Log:

[TIME_6] Attempt 6 (SUCCESSFUL): Executing reconnaissance commands

DESKTOP-91DJV8I\vbox

Victim's Log (Windows Event Viewer):

Event ID 4624 (Successful Logon):

Event ID: 4624

Computer: DESKTOP-91DJV8I

Account Name: vbox

Source Network Address: 192.168.56.103

Logon Type: 3 (Network - SSH)

Process Name: C:\Program Files\OpenSSH\sshd.exe

Status: 0x0 (Success)

Time: [TIMESTAMP_6]

PHASE 3: POST-COMPROMISE RECONNAISSANCE (T+13s to T+17s)

Attack Commands (From Kali):

sshpass -p "[PASSWORD]" ssh -l vbox 192.168.56.102 "whoami"

sshpass -p "[PASSWORD]" ssh -l vbox 192.168.56.102 "net user"

sshpass -p "[PASSWORD]" ssh -l vbox 192.168.56.102 "ipconfig"

Attacker's Log:

Execution of reconnaissance commands returned:

1. whoami → DESKTOP-91DJV8I\vbox (confirms identity)

2. net user → Lists all user accounts on system

3. ipconfig → Displays network configuration

Victim's Logs (Windows Event Viewer):

Event ID 4688 (Process Creation) - Entries 1-3:

Event ID: 4688

Process Command Line: C:\Windows\System32\cmd.exe /c whoami

Creator Process Name: C:\Program Files\OpenSSH\sshd.exe

Target User Name: vbox

Time: [TIMESTAMP_7]

Event ID: 4688

Process Command Line: C:\Windows\System32\cmd.exe /c net user

Creator Process Name: C:\Program Files\OpenSSH\sshd.exe

Target User Name: vbox

Time: [TIMESTAMP_8]

Event ID: 4688

Process Command Line: C:\Windows\System32\cmd.exe /c ipconfig

Creator Process Name: C:\Program Files\OpenSSH\sshd.exe

Target User Name: vbox

Time: [TIMESTAMP_9]

DETECTION ANALYSIS

INDICATORS OF COMPROMISE (IOCs)

1. BRUTE FORCE PATTERN (Event ID 4625)

What Blue Team Sees:

- 5 failed login attempts for same account (vbox)

- All from same source IP (192.168.56.103)

- Rapid succession (within seconds)
- Consistent failure reason: "Unknown user name or bad password"

Severity: 🟠 HIGH - Classic brute force signature

Recommended Action:
- Block source IP immediately
- Lock account for 30 minutes
- Alert security team

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2. SUCCESSFUL LOGIN AFTER FAILURES (Event ID 4624)

What Blue Team Sees:
- Successful login immediately after multiple failures
- Same source IP (192.168.56.103)
- Same account (vbox)
- Different status (0x0 = success vs 0xC000006D = failure)

Severity: 🟠 CRITICAL - Indicates compromise after brute force

Recommended Action:
- Account has been compromised
- Initiate incident response
- Preserve logs
- Isolate system if possible

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3. RECONNAISSANCE COMMANDS (Event ID 4688)

What Blue Team Sees:

- Commands executed immediately after successful login
- Execution pattern shows system enumeration:
  - whoami - Identity confirmation
  - net user - User enumeration
  - ipconfig - Network reconnaissance

Severity: 🔴 CRITICAL - Active attacker on system

Attack Intent:

1. Confirmation - Verify compromised identity
2. Enumeration - Map system users and groups
3. Network Mapping - Identify network configuration
4. Preparation - Gathering info for privilege escalation

Recommended Action:

- IMMEDIATE: Isolate system from network
- Review what commands will follow
- Backup logs for forensics
- Begin incident response investigation

CROSS-PLATFORM CORRELATION

TIMELINE CORRELATION

Kali (Attacker) View:

19:45:30 - Brute force script started

19:45:32 - Attempt 1: wrongpass1 (Permission denied)

19:45:34 - Attempt 2: wrongpass2 (Permission denied)

19:45:36 - Attempt 3: wrongpass3 (Permission denied)

19:45:38 - Attempt 4: wrongpass4 (Permission denied)

19:45:40 - Attempt 5: wrongpass5 (Permission denied)

19:45:42 - Attempt 6: SUCCESSFUL (vbox account compromised)

19:45:43 - Command: whoami (Identity confirmed)

19:45:45 - Command: net user (Users enumerated)

19:45:47 - Command: ipconfig (Network mapped)

Windows (Victim) View:

19:45:32 - Event 4625: Failed login attempt 1

19:45:34 - Event 4625: Failed login attempt 2

19:45:36 - Event 4625: Failed login attempt 3

19:45:38 - Event 4625: Failed login attempt 4

19:45:40 - Event 4625: Failed login attempt 5

19:45:42 - Event 4624: SUCCESSFUL login (vbox, 192.168.56.103)

19:45:43 - Event 4688: whoami.exe executed

19:45:45 - Event 4688: net.exe executed

19:45:47 - Event 4688: ipconfig.exe executed

Correlation Result:

✓ Perfect alignment between attacker and victim timelines

✓ Both sources confirm same attack sequence

✓ Cross-platform evidence validates incident

SECURITY FINDINGS

1. ACCOUNT COMPROMISE VIA BRUTE FORCE

Finding: The vbox user account was successfully compromised after 5 password guessing attempts.

Root Cause:

- Weak password policy (no account lockout)
- No multi-factor authentication
- Account allowed unlimited failed login attempts

Evidence:

- 5x Event ID 4625 (failures)
- 1x Event ID 4624 (success after failures)

Severity: CRITICAL

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2. INSUFFICIENT ACCESS CONTROLS

Finding: SSH service allows remote password-based authentication without restrictions.

Evidence:

- Multiple failed login attempts accepted
- No temporary blocking after N failures
- No rate limiting on authentication attempts

Severity: 🟠 HIGH

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3. POST-COMPROMISE RECONNAISSANCE

Finding: Attacker executed system enumeration commands immediately after gaining access.

Evidence:
- 3x Event ID 4688 (process execution)
- Commands: whoami, net user, ipconfig
- All executed within 5 seconds of successful login

Severity: 🟠 CRITICAL - Indicates active attacker mapping system

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RECOMMENDATIONS

IMMEDIATE ACTIONS (Do Now)

1. 🟠 RESET vbox ACCOUNT PASSWORD
   Impact: Medium (user will need new password)
   Risk: Mitigates account compromise

2. 🟠 BLOCK SOURCE IP 192.168.56.103

Windows Firewall Rule:

netsh advfirewall firewall add rule name="Block Attacker"

dir=in action=block remoteip=192.168.56.103

3. 🟢 REVIEW ALL ACTIVITY BY vbox ACCOUNT

- Check Event Viewer for other suspicious events
- Review file access logs
- Check for privilege escalation attempts

SHORT-TERM ACTIONS (This Week)

1. ENABLE ACCOUNT LOCKOUT POLICY

Action: Lock account after 5 failed attempts

Duration: 30 minutes

Method: Local Security Policy or GPO

2. IMPLEMENT MULTI-FACTOR AUTHENTICATION

Requirement: SSH key-based auth instead of passwords

OR: Time-based one-time password (TOTP)

3. ENABLE COMPREHENSIVE LOGGING

- Event ID 4625 (failed logins) - ✓ Currently enabled
- Event ID 4624 (successful logins) - ✓ Currently enabled
- Event ID 4688 (process execution) - ✓ Currently enabled
- Keep logs for minimum 90 days

LONG-TERM ACTIONS (Next Month)

1. DISABLE SSH PASSWORD AUTHENTICATION

sshd_config:

PasswordAuthentication no

PubkeyAuthentication yes

2. IMPLEMENT INTRUSION DETECTION

- Deploy IDS/IPS
- Monitor for brute force patterns
- Automatic blocking of sources exceeding threshold

3. SECURITY AWARENESS TRAINING

- Teach users about strong passwords
- Phishing and social engineering awareness
- How to report suspicious activity

4. REGULAR SECURITY ASSESSMENTS

- Penetration testing
- Vulnerability scanning
- Log review procedures

TECHNICAL DETAILS

WINDOWS EVENT IDS EXPLAINED

Event ID 4625 (Audit Failure - Logon)

Generated when: Login attempt fails

Logged on: Computer being attacked

Shows: Failed authentication details

Key fields: Account name, source IP, failure reason

Event ID 4624 (Audit Success - Logon)

Generated when: Login attempt succeeds

Logged on: Computer being attacked

Shows: Successful authentication details

Key fields: Account name, source IP, logon type

Event ID 4688 (Process Creation)

Generated when: New process is created

Logged on: Computer where process runs

Shows: Process details and command line

Key fields: Process name, creator process, command line

LOGON TYPES

Type	Name	Used For	Security Risk
2	Interactive	Local keyboard login	Low (physical)
3	Network	SSH, RDP, SMB, network	High (remote)
4	Batch	Scheduled tasks	Medium (automation)
5	Service	Service account login	High (privilege)
10	RemoteInteractive	RDP/Remote Desktop	High (remote)

In this lab: Logon Type 3 (Network) = SSH remote access

LAB ENVIRONMENT

Attacker System:

- OS: Kali Linux

- IP: 192.168.56.103

- Tools Used: sshpass, ssh client, bash scripting

Victim System:

- OS: Windows 11

- IP: 192.168.56.102

- Service: OpenSSH Server (port 22)

- Logging: Windows Security Event Log

Network:

- Type: Host-only network (VMware)

- Isolation: Lab only, no internet access

- Purpose: Safe simulation environment

LESSONS LEARNED

FOR BLUE TEAM (DEFENDERS)

1. ✓ Brute force attacks leave clear evidence in logs

2. ✓ Failed and successful logins can be correlated

3. ✓ Process execution logs show attacker actions

4. ✓ Cross-platform logging provides complete picture

5. ✓ Account lockout policies are essential

6. ✓ Real-time alerting prevents full compromise

FOR RED TEAM (ATTACKERS)

1. ✓ Brute force works but creates detectable patterns

2. ✓ Attacker commands are logged immediately

3. ✓ Post-compromise reconnaissance is tracked

4. ✓ Both attacker and victim logs must be considered

5. ✓ Stealthy attacks require log tampering (advanced)

FOR PURPLE TEAM

1. ✓ Understanding both perspectives enables better defense

2. ✓ Attack timeline reconstruction is critical

3. ✓ Log correlation reveals complete incident

4. ✓ Rapid response can contain ongoing attack

5. ✓ Proactive measures (lockout, MFA) prevent success

FILES INCLUDED

- ssh_attack_log.txt - Attacker's command execution log

- ssh_command_history.txt - Kali bash history (SSH commands)

- windows_event_4625.txt - Failed login events

- windows_event_4624.txt - Successful login events

- windows_event_4688.txt - Process execution events

- SSH_Brute_Force_Lab_Report.md - This report

- README.md - Lab overview and setup instructions

CONCLUSION

This lab successfully demonstrated a complete attack lifecycle from initial access through post-compromise reconnaissance. Detection was achieved through proper logging configuration and log analysis, showing that well-configured systems can detect and respond to intrusions effectively.

The cross-platform analysis (Kali + Windows) demonstrates that attackers leave evidence on multiple systems, and comprehensive logging and analysis is essential for modern security operations.

Lab Created: [YOUR_DATE]

Duration: ~30 seconds (attack execution)

Lab Setup Time: ~2 hours

Documentation Time: ~1 hour

Total Lab Time: ~3 hours

ABOUT THIS LAB

This lab was created as part of a Purple Team Cybersecurity Training Program, demonstrating:

- Attack simulation techniques
- Cross-platform log analysis
- Incident detection capabilities
- Professional documentation practices

FOR EDUCATION & AUTHORIZED SECURITY TESTING ONLY
